FORDHAM
THE JESUIT UNIVERSITY OF NEW YORK

Gabelli School of Business

Lecture III – Decision Trees

Machine Learning for Business

Michael Deamer

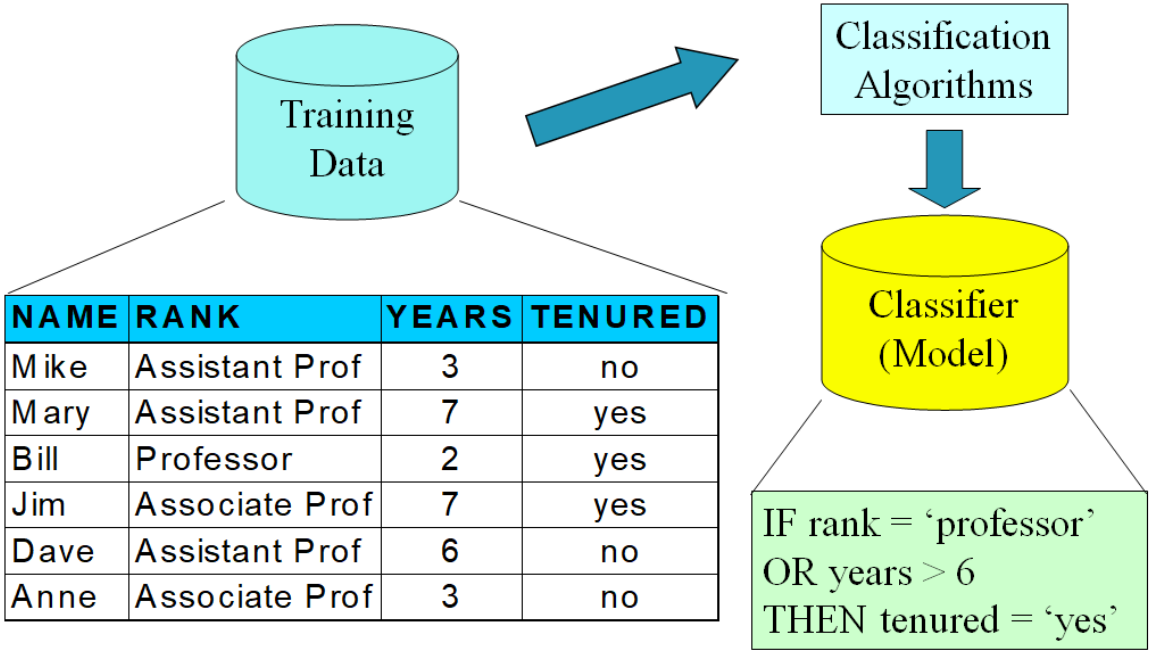
Classification

Prediction

Surface	Mouth	Nose	Voice	Locomotion	Class
Skin	Oval	Snout	Squeek	Walk	Pig
Feathers	Bill	Holes	Quack	Waddle	Duck
Skin	Oval	Aquiline	Twitter	Automobile	Human
Feathers	Bill	Holes	Quack	Swims	?

Classification Process

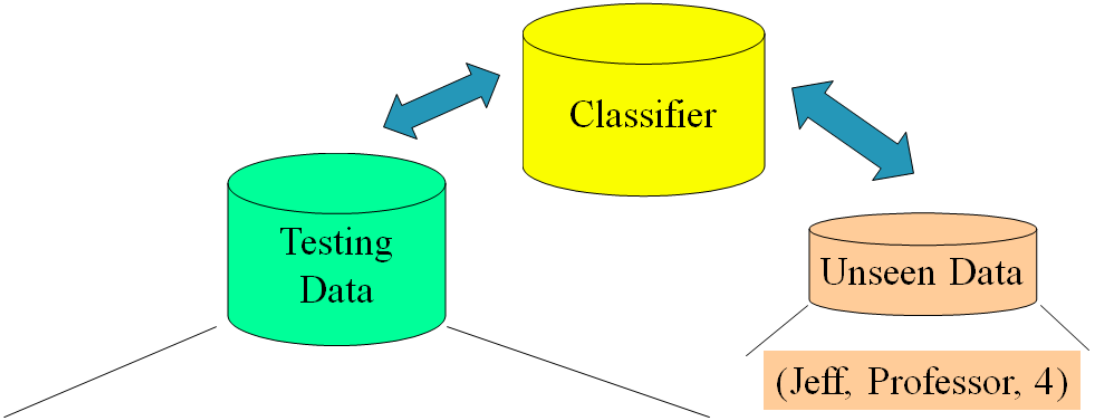
Step I – Model Construction



NAME	RANK	YEARS	TENURED
Mike	Assistant Prof	3	no
Mary	Assistant Prof	7	yes
Bill	Professor	2	yes
Jim	Associate Prof	7	yes
Dave	Assistant Prof	6	no
Anne	Associate Prof	3	no

Classification Process

Step II – Test Accuracy of Model



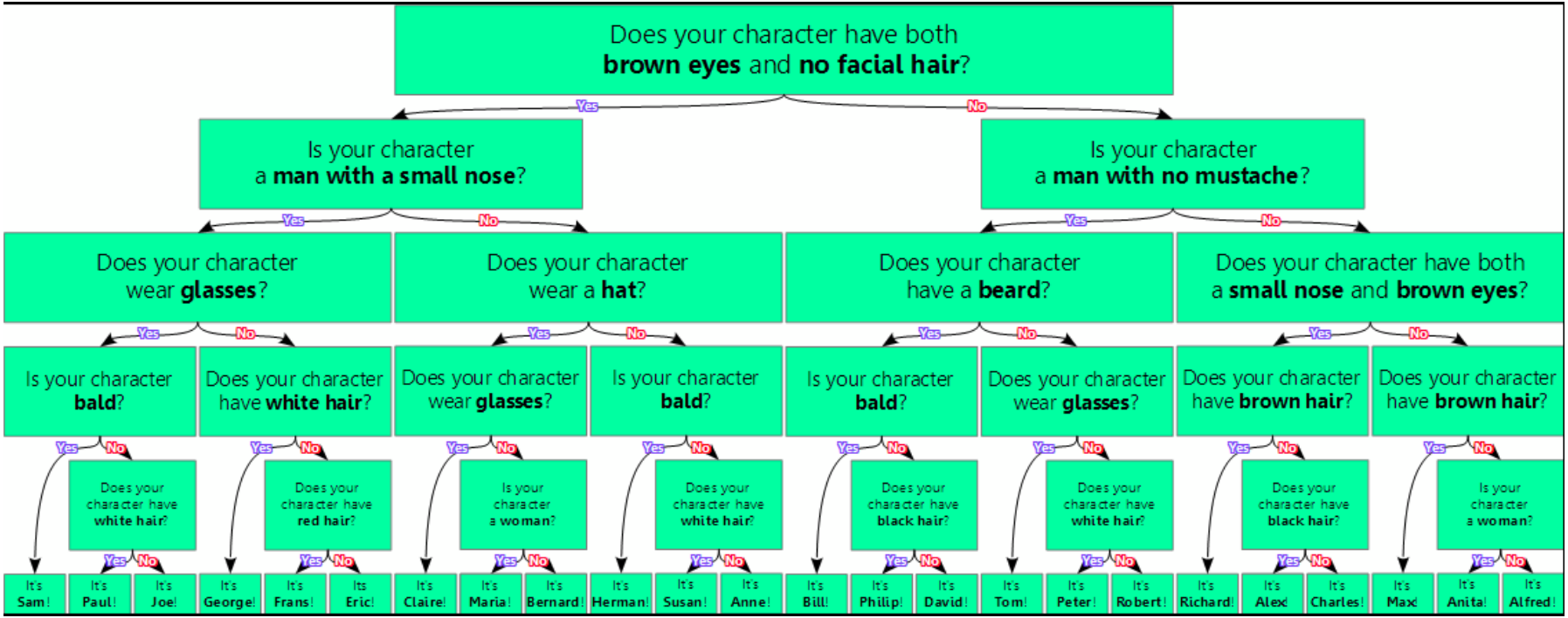
NAME	RANK	YEARS	TENURED
Tom	Assistant Prof	2	no
Merlisa	Associate Prof	7	no
George	Professor	5	yes
Joseph	Assistant Prof	7	yes

Example

Guess who

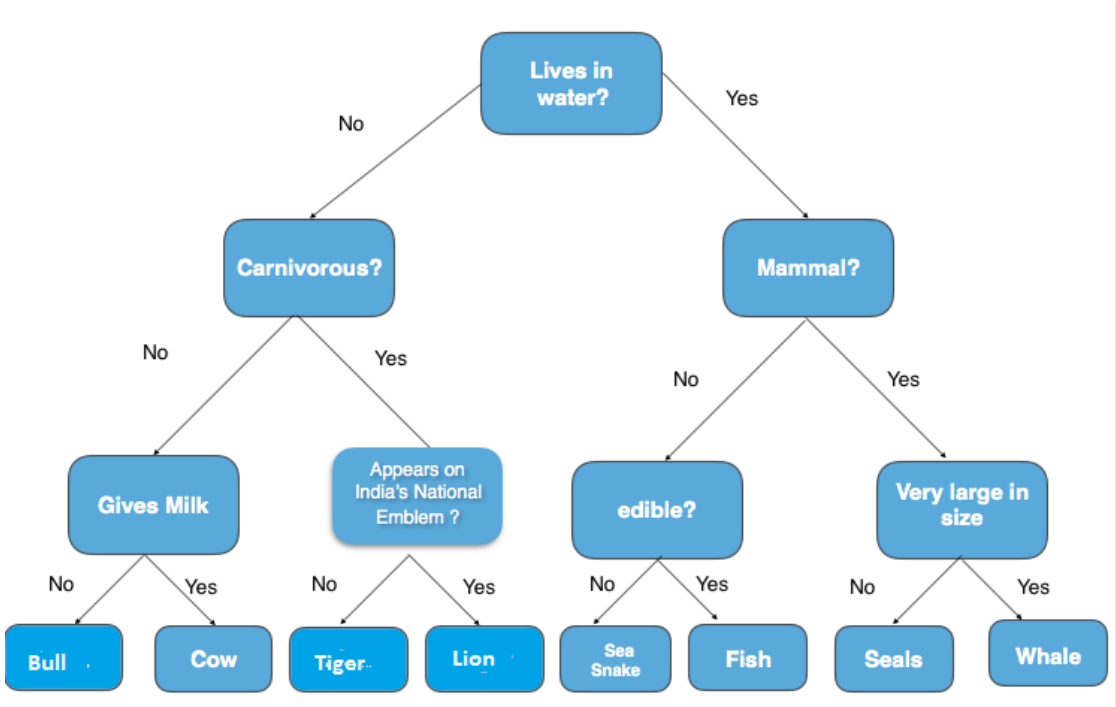


Example con't



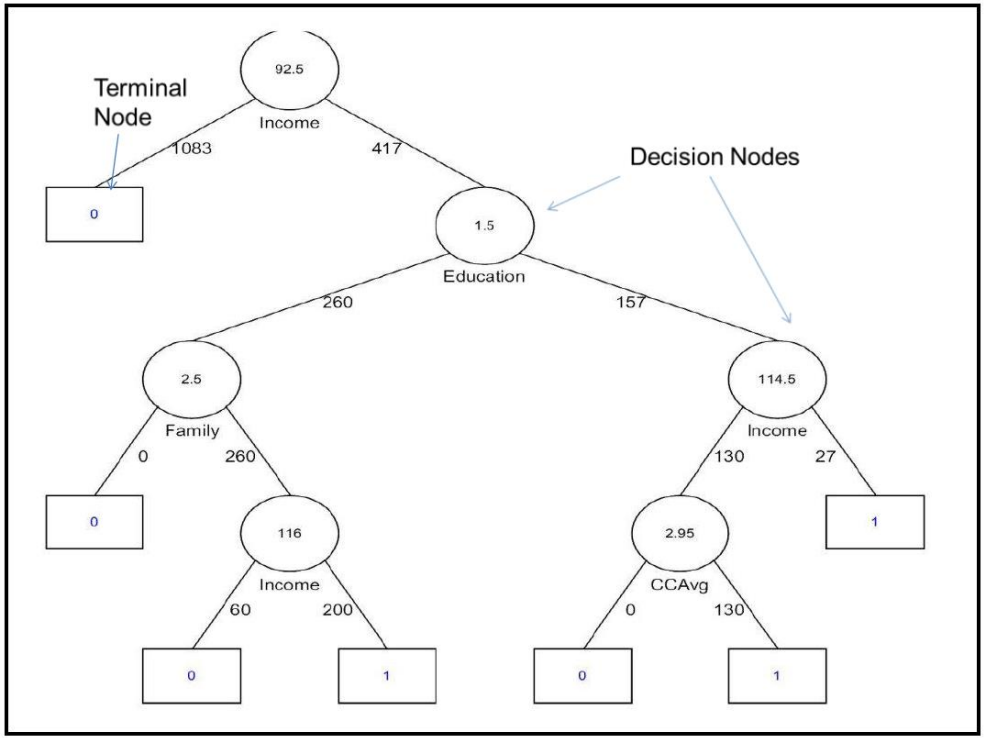
Example 2

Animal

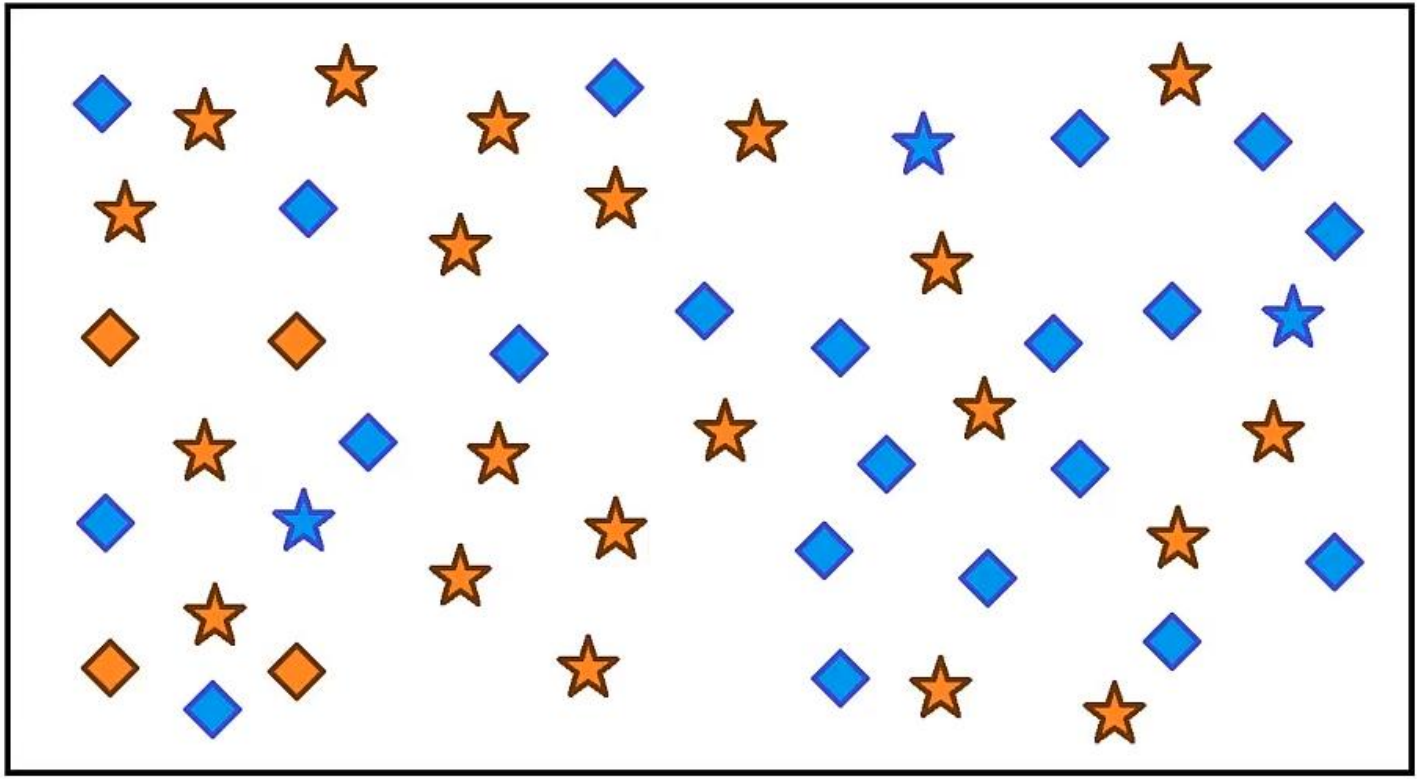


Example 3

Non-binary fields



Entropy



Entropy

$$E = -\sum_{x=1}^{n} p_x \bullet \log_2 p_x$$

$$E = -(p1 \log_2 p1 + p2 \log_2 p2)$$

Entropy

Full Group:

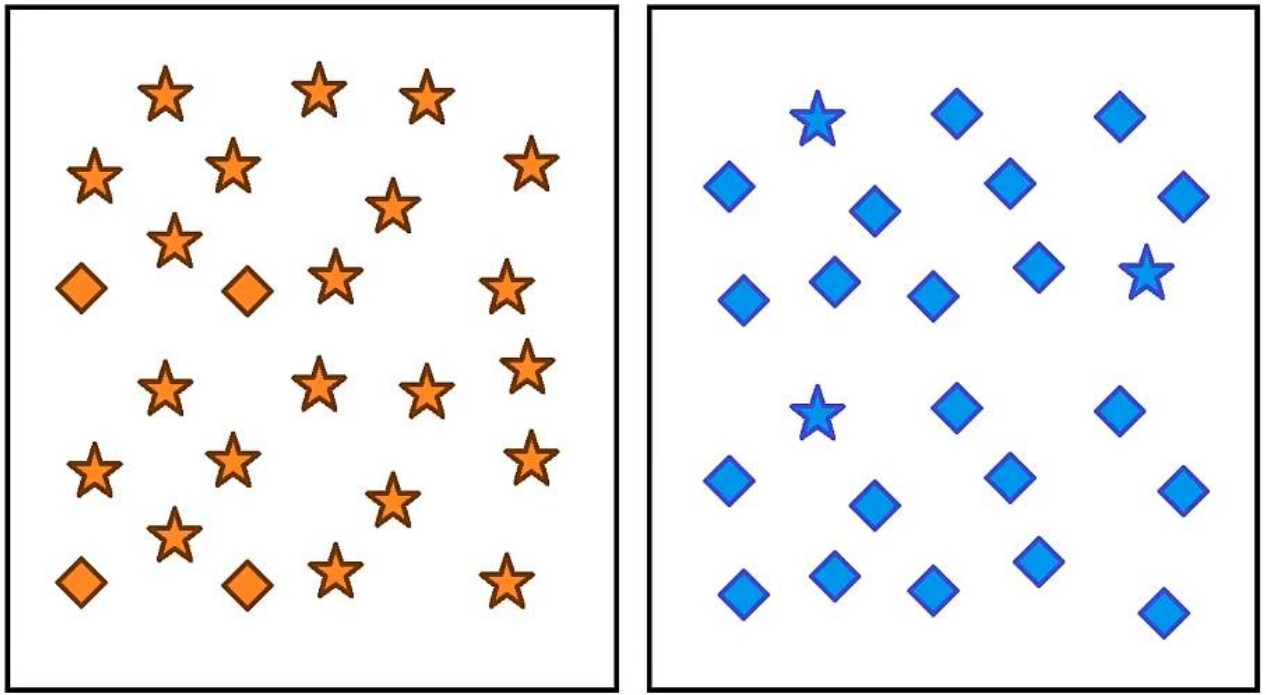
$$E(\vec{d}) = -\sum_i p_i \log_2(p_i)$$

$$= -(p_1 \log_2(p_1) + p_2 \log_2(p_2))$$

$$= -\left(\frac{24}{49}\log_2\left(\frac{24}{49}\right) + \frac{25}{49}\log_2\left(\frac{25}{49}\right)\right)$$

$$\cong 0.9997$$

Entropy



Entropy

Orange Group:

$$E(\vec{d}, orange) = -\sum_i p_i \log_2(p_i)$$

$$= -(p_1 \log_2(p_1) + p_2 \log_2(p_2))$$

$$= -\left(\frac{4}{25}\log_2\left(\frac{4}{25}\right) + \frac{21}{25}\log_2\left(\frac{21}{25}\right)\right)$$

$$\cong 0.6343$$

Entropy

Blue Group:

$$E(\vec{d}, blue) = -\sum_i p_i \log_2(p_i)$$

$$= -(p_1 \log_2(p_1) + p_2 \log_2(p_2))$$

$$= -\left(\frac{21}{24}\log_2\left(\frac{21}{24}\right) + \frac{3}{24}\log_2\left(\frac{3}{24}\right)\right)$$

$$\cong 0.5436$$

Entropy

Combined Entropy:

$$E(\vec{d}) = 0.9997$$

$$E(\vec{d}, \vec{a}) = \frac{25}{49} E(orange) + \frac{24}{49} E(blue)$$

$$= \frac{25}{49}(0.6343) + \frac{24}{49}(0.5436) = 0.5899$$

Entropy

Combined Entropy:

$$E(\vec{d}) = 0.9997$$

$$E(\vec{d}, \vec{a}) = \frac{25}{49} E(orange) + \frac{24}{49} E(blue)$$

$$= \frac{25}{49}(0.6343) + \frac{24}{49}(0.5436) = 0.5899$$

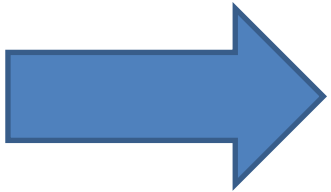
Information Gain:

$$I(\vec{d}, \vec{a}) = E(\vec{d}) - E(\vec{d}, \vec{a}) = 0.9997 - 0.5899 = 0.4097$$

Continuous Fields – Lawn Mower

Step 1: Order the field

Income	Lot_Size	Ownership
60.0	18.4	owner
85.5	16.8	owner
64.8	21.6	owner
61.5	20.8	owner
87.0	23.6	owner
110.1	19.2	owner
108.0	17.6	owner
82.8	22.4	owner
69.0	20.0	owner
93.0	20.8	owner
51.0	22.0	owner
81.0	20.0	owner
75.0	19.6	non-owner
52.8	20.8	non-owner
64.8	17.2	non-owner
43.2	20.4	non-owner
84.0	17.6	non-owner
49.2	17.6	non-owner
59.4	16.0	non-owner
66.0	18.4	non-owner
47.4	16.4	non-owner
33.0	18.8	non-owner
51.0	14.0	non-owner
63.0	14.8	non-owner

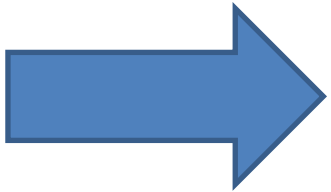


Income	LotSize	Owner
51	14	No
63	14.8	No
59.4	16	No
47.4	16.4	No
85.5	16.8	Yes
64.8	17.2	No
108	17.6	Yes
84	17.6	No
49.25	17.6	No
60	18.4	Yes
66	18.4	No
33	18.8	No
110.1	19.2	Yes
75	19.6	No
69	20	Yes
81	20	No
43.2	20.4	No
61.5	20.8	Yes
93	20.8	No
52.8	20.8	No
64.8	21.6	Yes
51	22	No
82.8	22.4	Yes
87	23.6	Yes

Continuous Fields – Lawn Mower

Step 2: Calculate midpoints

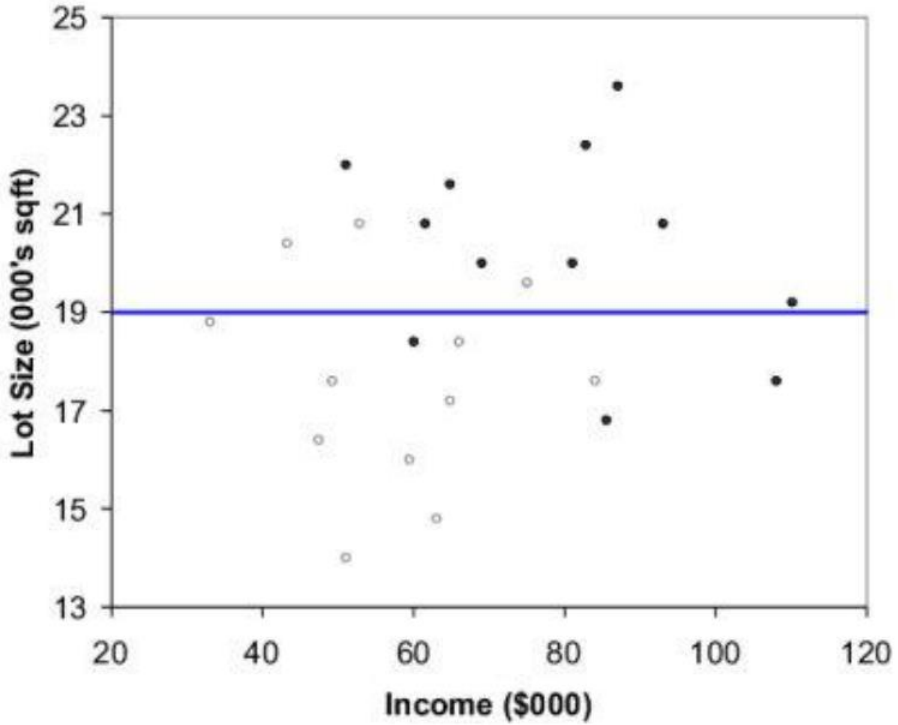
Income	LotSize	Owner
51	14	No
63	14.8	No
59.4	16	No
47.4	16.4	No
85.5	16.8	Yes
64.8	17.2	No
108	17.6	Yes
84	17.6	No
49.25	17.6	No
60	18.4	Yes
66	18.4	No
33	18.8	No
110.1	19.2	Yes
75	19.6	No
69	20	Yes
81	20	No
43.2	20.4	No
61.5	20.8	Yes
93	20.8	No
52.8	20.8	No
64.8	21.6	Yes
51	22	No
82.8	22.4	Yes
87	23.6	Yes



Income	LotSize	Owner	
51	14	No	14.4
63	14.8	No	15.4
59.4	16	No	16.2
47.4	16.4	No	16.6
85.5	16.8	Yes	17
64.8	17.2	No	17.4
108	17.6	Yes	17.6
84	17.6	No	17.6
49.25	17.6	No	18
60	18.4	Yes	18.4
66	18.4	No	18.6
33	18.8	No	19
110.1	19.2	Yes	19.4
75	19.6	No	19.8
69	20	Yes	20
81	20	No	20.2
43.2	20.4	No	20.6
61.5	20.8	Yes	20.8
93	20.8	No	20.8
52.8	20.8	No	21.2
64.8	21.6	Yes	21.8
51	22	No	22.2
82.8	22.4	Yes	23
87	23.6	Yes	11.8

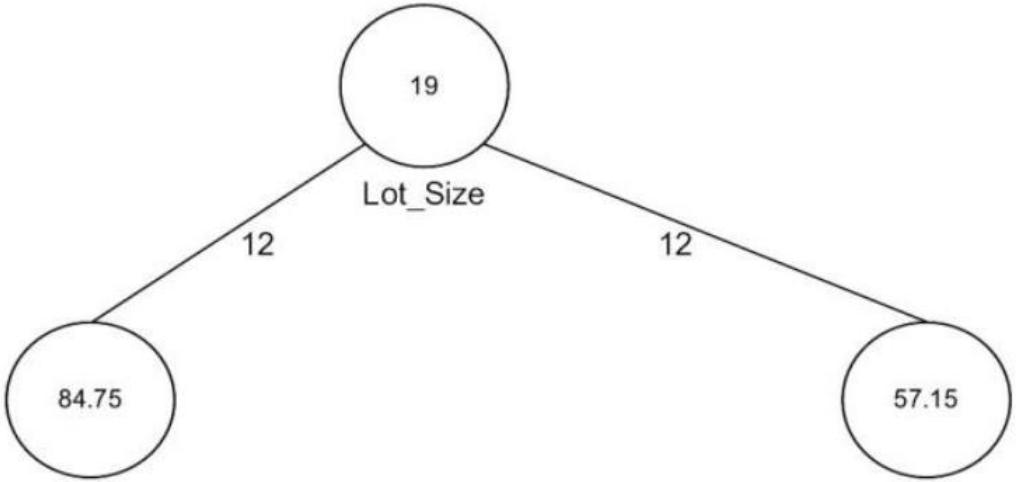
Continuous Fields – Lawn Mower

Step 3: Split on each midpoint, evaluate, and repeat



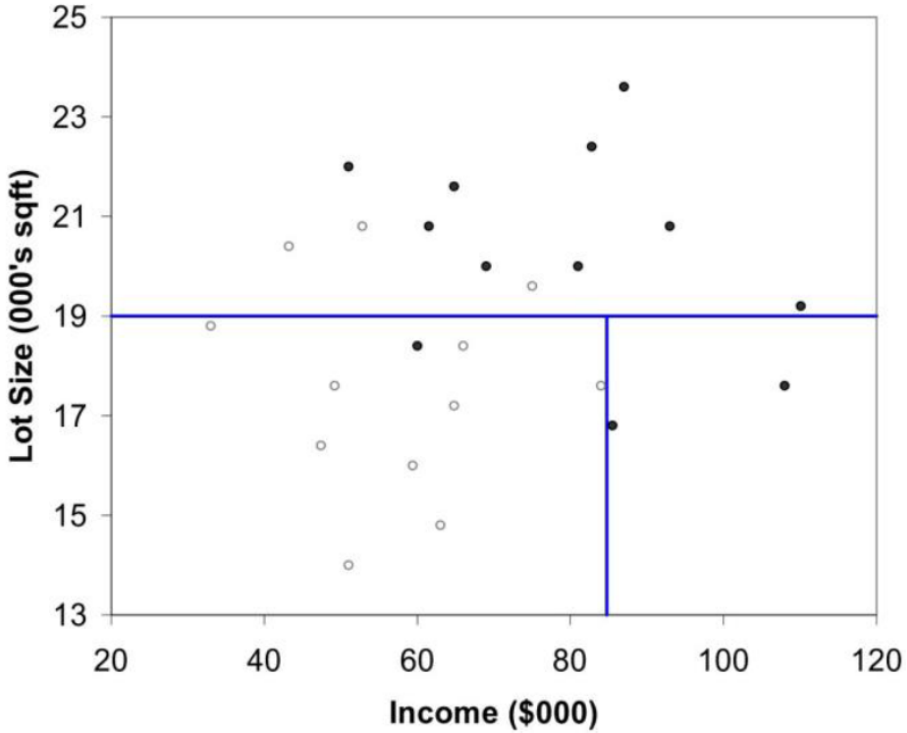
Continuous Fields – Lawn Mower

Step 4: Branch tree at optimal split point



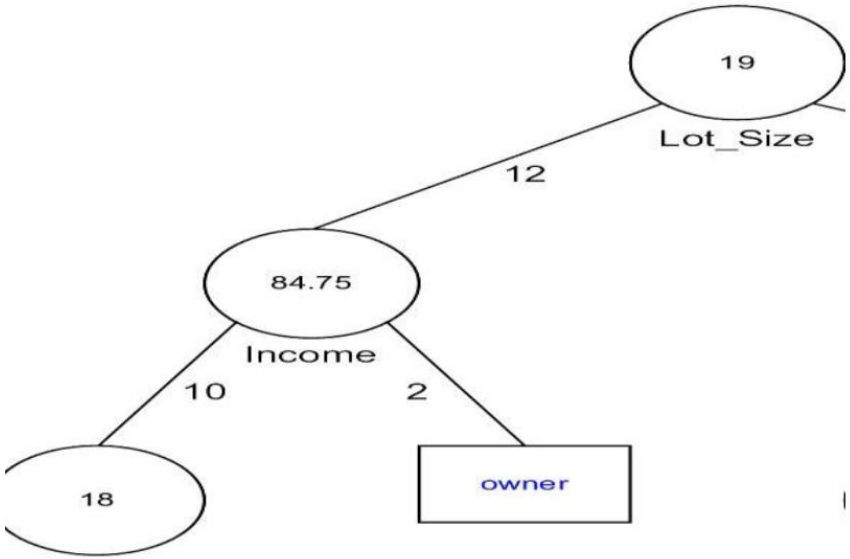
Continuous Fields – Lawn Mower

Step 5: Repeat for other fields



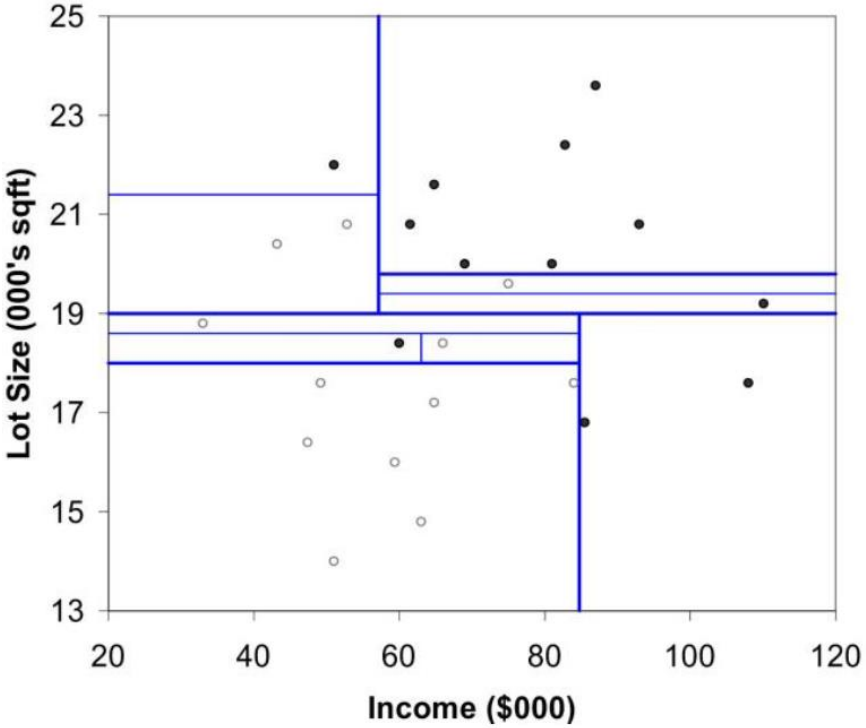
Continuous Fields – Lawn Mower

Step 4: Branch tree at optimal split point



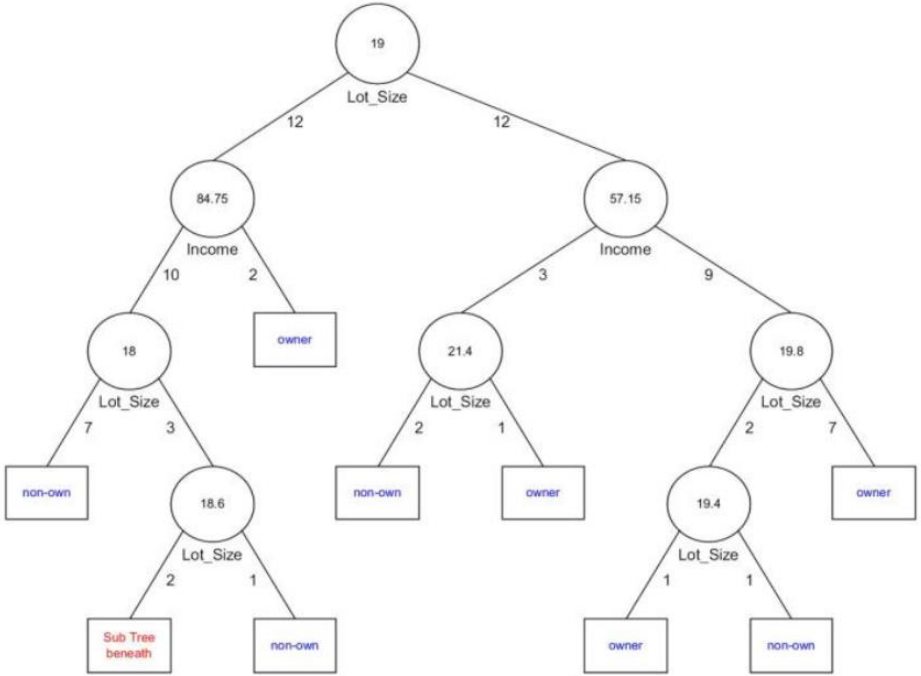
Continuous Fields – Lawn Mower

Result



Continuous Fields – Lawn Mower

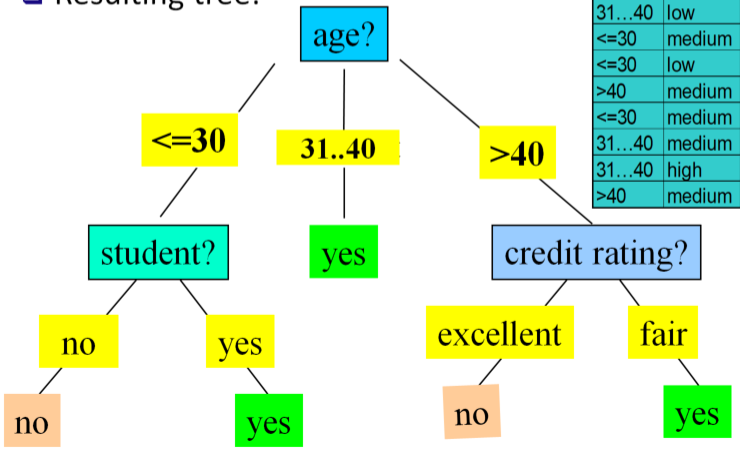
Result Tree



Splits aren't always binary

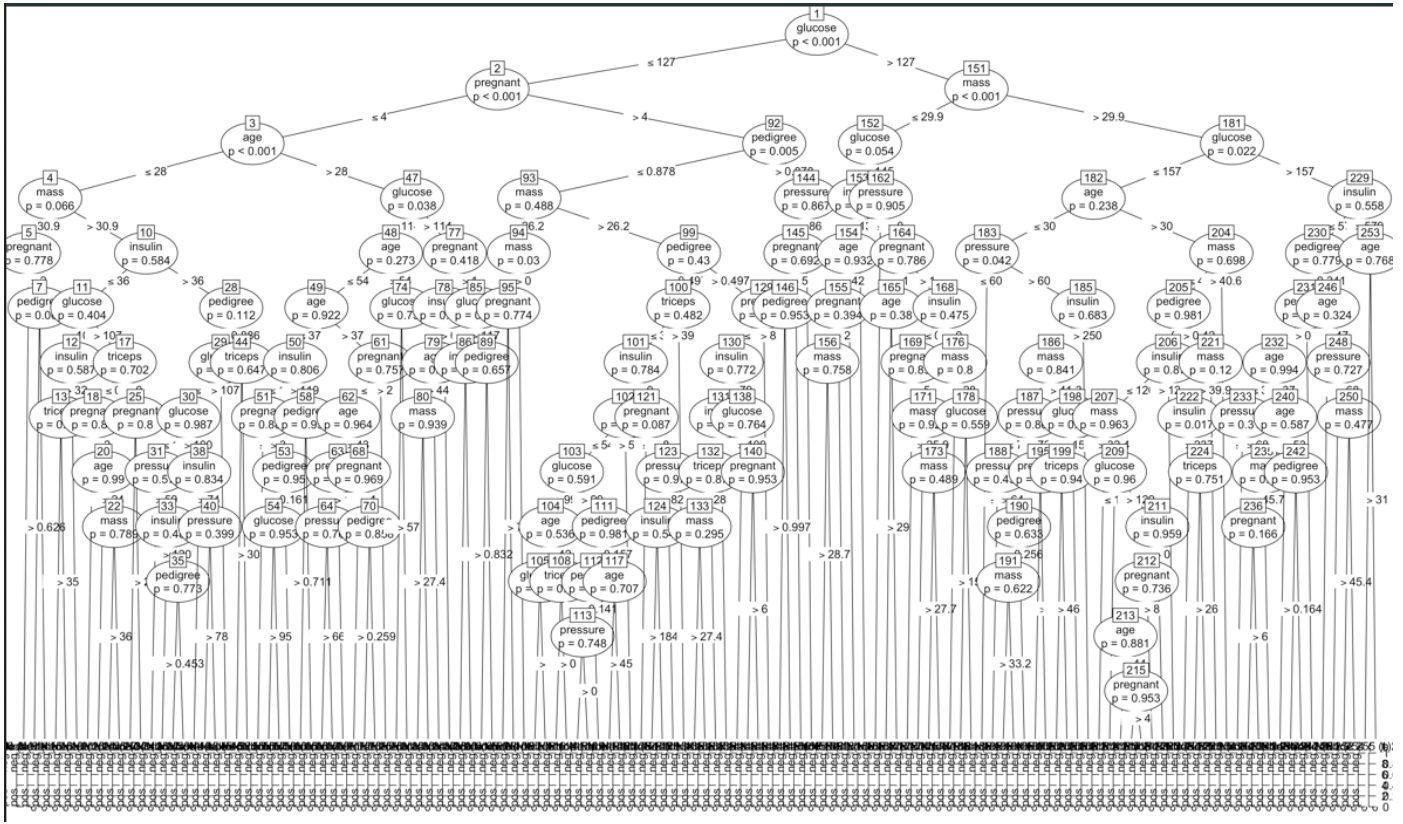
Decision Tree Induction: An Example

- Training data set: Buys_computer
- The data set follows an example of Quinlan's ID3 (Playing Tennis)
- Resulting tree:



age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

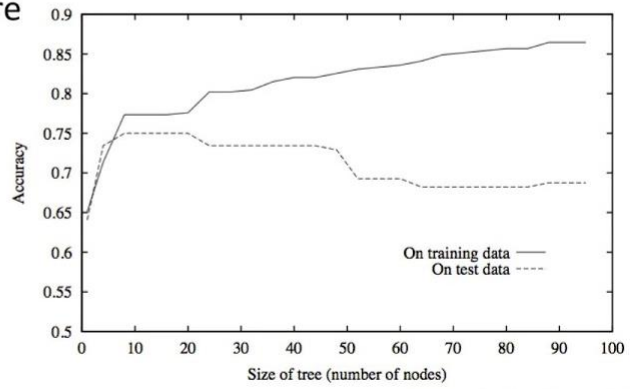
Overfitting



Identifying Overfitting

Overfitting in Decision Trees

- Can always classify training examples perfectly
  - keep splitting until each node contains 1 example
  - singleton = pure
- Doesn't work on new data

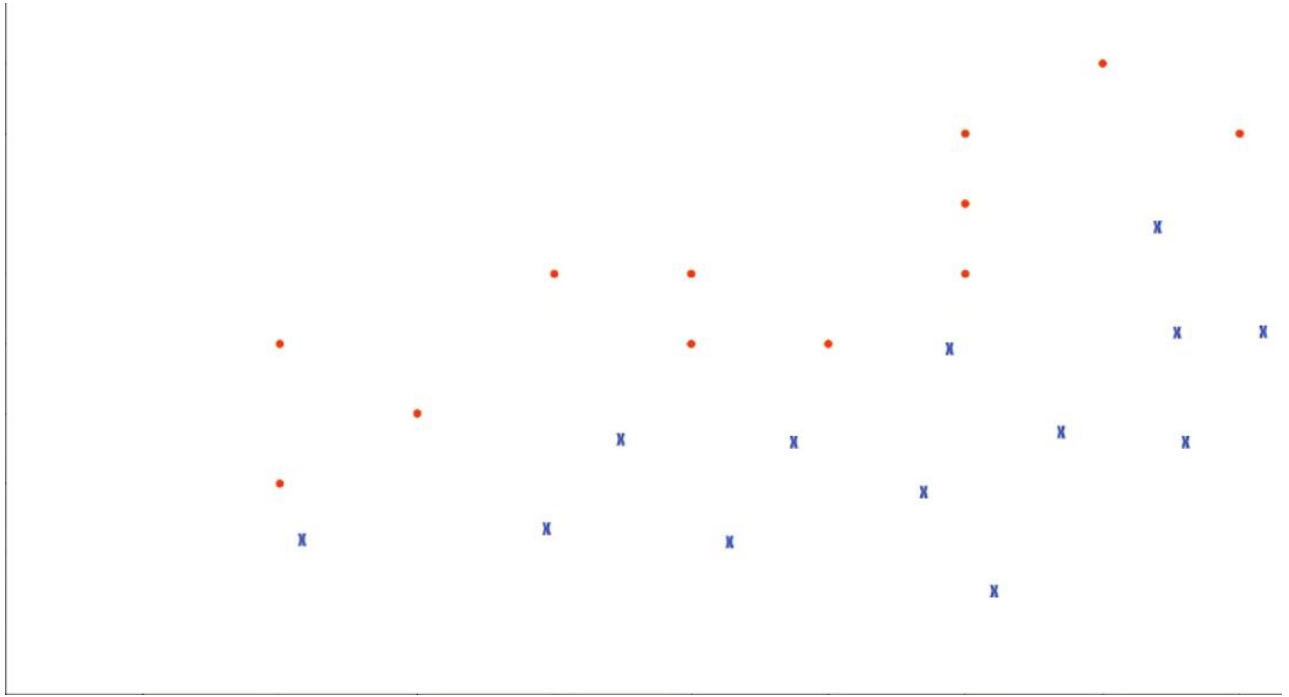


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Figure credit: Tom Mitchell, 1997

Limitation

Interaction between variables



To Do

- ❑ Lab II
- ❑ Homework II
- ❑ Connect with group via email and schedule regular meetings
- ❑ Recommended Reading: 6.6 and 6.7

Bibliography

Tan, Steinbach, Karpatne, and Kumar. Introduction to Data Mining, second edition. Pearson, Hudson, NY, 2019.
Han, Kamber, Pei. Data Mining Concepts and Techniques, second edition. University of Illinois at Urbana-Champaign. Elsevier, San Francisco, CA, 2006.